The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

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October 28, 2026

1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Second-Order Kuramoto Grid Physics and Decentralized CSMA/CD Access

We formalize the active stabilization of shared electrical transmission networks under massive distributed training workloads as a closed-loop system mapping discrete multi-agent reinforcement learning to continuous swing equations.

3.1 The Continuous Route to Chaos Swing Matrix

Let N represent the total number of synchronized training nodes scaling asymptotically up to $10,000$. Each electrical substation bus i behaves as a coupled non-linear oscillator with an instantaneous phase angle $\theta_i(t)$ and a local phase velocity deviation $\dot{\theta}_i(t)$. The physical grid dynamics under the impact of periodic data center loads are governed by the second-order Kuramoto swing equation:

$$m_i \ddot{\theta}_i + d_i \dot{\theta}_i = P_i - \sum_{j=1}^{N} K_{ij} \sin(\theta_i - \theta_j) - P_{\text{AI},i}(t) \tag{2}$$

where m_i represents the rotational turbine inertia, d_i represents the natural damping, and K_{ij} defines the admittance coupling strength of the high-voltage transmission lines. The term $P_{\text{AI},i}(t)$ represents the aggregate load injection of the AI cluster. Prior to reinforcement learning intervention, uncoordinated training sessions run in a perfectly synchronous software loop, transforming $P_{\text{AI},i}(t)$ into a heavy low-frequency square wave pulsing at a destructive resonant frequency:

$$P_{\text{AI},i}(t) = \text{Square_Wave}(\Omega_{\text{load}} \cdot t + \phi_i), \quad \Omega_{\text{load}} \approx 0.5 \text{ Hz} \tag{3}$$

As the amplitude of this periodic driving force scales up, the grid frequency deviations $\dot{\theta}_i(t)$ undergo a classic period-doubling bifurcation cascade, collapsing the global phase-locked limit cycle into a chaotic attractor, causing extensive phase-slipping and localized grid blackouts.

3.2 Multi-Facility Scaling and Normalized Sensing Operators

As the architecture scales up to 10,000 training nodes distributed across multiple locations at a density of 50 training nodes per facility, static global threshold sensing breaks down. The MARL agents adapt by implementing a normalized carrier-sensing metric $\overline{L}_i(t)$ tracking local facility capacity perimeters:

$$\overline{L}_i(t) = \frac{\sum_{j \in \mathcal{N}_i} P_{\text{AI},j}(t)}{\text{Capacity}_{\text{facility}}} \tag{4}$$

This multi-scale localization allows individual clusters to sense the aggregate load acceleration ($\frac{dL}{dt}$) and execute independent out-of-phase gating delays, preventing the formation of localized chaotic attractors.

3.3 Phase-Shift Stabilization Under Rigid Periodic Software Forcing

To prevent grid collapse without intruding into the proprietary data center software stack, the MARL controller acts exclusively as an external access gatekeeper at the network boundary interface. Let $\mathcal{H}_{\text{CD}}$ represent the collision detection function tracking aggregate load derivatives ($\frac{dL}{dt}$). Upon detection of an imminent bus overlap event, the controller instantiates a randomized temporal phase-shifting gating operator $\mathcal{G}_{\Delta t}$:

$$P_{\text{AI},i}^{\text{gated}}(t) = \text{Square_Wave}\Big(\Omega_{\text{load}} \cdot (t - \Delta t_{\text{random}}) + \phi_i\Big) \tag{5}$$

The application of $\mathcal{G}_{\Delta t}$ introduces an artificial, out-of-band asymmetric delay that forces destructive interference across the active load profiles:

$$\phi_1 \neq \phi_2^{\text{gated}} \implies \int_t^{t+T} \Big(P_{\text{AI},1}(t) + P_{\text{AI},2}^{\text{gated}}(t)\Big) e^{-i\Omega_{\text{load}} t} dt \to 0 \tag{6}$$

By explicitly nullifying the low-frequency Fourier spectral magnitude within the critical resonant inter-area band ($0.1 - 20$ Hz), the destructive driving term is removed from the Kuramoto system manifold.

Following the synchronized chaos boundary criteria first established in the non-linear dynamics research of **de Sousa Vieira, Sherman, et al. (1996)**, removing this periodic forcing profile allows the grid's natural damping d_i and mechanical turbine inertia m_i to recover immediate dominance. The frequency deviations $\dot{\theta}_i(t)$ decay exponentially, pulling the coupled non-linear oscillators back into a stable, globally synchronized phase-locked steady state ($\circ$).

Our inductive procedures validate that the emergent power grid utility service curve strictly satisfies standard network performance curves for throughput (S) versus offered traffic load (G) first established in the local network implementation benchmarks of **Sherman & Gable (1983)**.